

Validation of 2D Supersonic Nozzle Flow Against 1D Isentropic Theory

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Abstract

This study validates a two-dimensional, density-based CFD simulation of a supersonic converging-diverging nozzle against theoretical 1D isentropic gas dynamics to ensure numerical reliability for propulsion applications. Using a structured quadrilateral mesh and a Roe-FDS flux-splitting scheme in Ansys Fluent, the simulation achieved grid independence with a 0.06% Mach number variation between refinement levels. Final validation results yielded an exit Mach number of 2.204, representing a 0.2% error from the 2.20 design target, and a mass flow rate of 11.714 kg/s with a negligible error. Additionally, the calculated net thrust of 5,297 N demonstrated a 1.1% deviation from the theoretical value of 5,358 N, confirming the model's ability to accurately capture momentum flux and over-expanded pressure effects. Qualitative analysis confirmed the resolution of 2D Prandtl-Meyer expansion waves, verifying the efficacy of the density-based solver setup for high-speed compressible flow analysis.

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Nomenclature

Symbol	Description
A	Local cross-sectional area
A^*	Throat area (area of minimum cross-section)
A_e	Nozzle exit area
F	Total net thrust
M	Local Mach number
M_e	Exit Mach number

$\dot{m}$	Mass flow rate
P	Static pressure
P_0	Stagnation (total) pressure
P_a	Ambient (atmospheric) pressure
P_e	Static pressure at the nozzle exit
R	Specific gas constant (287 for air)
T	Static temperature
T_0	Stagnation (total) temperature
T_e	Static temperature at the nozzle exit
V	Local flow velocity
V_e	Exit flow velocity
γ	Ratio of specific heats (1.4 for air)

1 Introduction

The development and optimization of high speed propulsion systems rely heavily on the accurate prediction of internal flow behaviors within supersonic nozzles. This project focuses on the numerical simulation and physical validation of a two dimensional converging-diverging nozzle designed for a specific expansion ratio. The problem statement involves the modeling of compressible fluid flow through a nozzle to verify that the numerical results match established theoretical gas dynamics. For a valid comparison between the computational results and the published isentropic data, certain flow properties and conditions must be maintained. These conditions include a constant stagnation pressure of 506,625 Pascals and a stagnation temperature of 305.6 Kelvin at the inlet, along with the requirement that the flow reaches a choked condition at the throat. The geometry specifically consists of a nozzle with a throat height of 0.01 meters and an exit height of 0.02 meters which yields a design area ratio of 2.0. According to isentropic flow theory, this specific area ratio should produce a target exit Mach number as defined by the following relation

$$\frac{A}{A^*} = \frac{1}{M} \left[\frac{2}{\gamma + 1} \left(1 + \frac{\gamma - 1}{2} M^2 \right) \right]^{\frac{\gamma + 1}{2(\gamma - 1)}}$$

For the specified area ratio of 2.0, the target exit Mach number for the validation is 2.2. By comparing the numerical outputs from Fluent to these theoretical predictions, this study aims to verify the accuracy of the density based solver settings and mesh configurations. This process ensures the computational model properly captures the transition from subsonic to supersonic regimes as shown in the geometry schematic below.

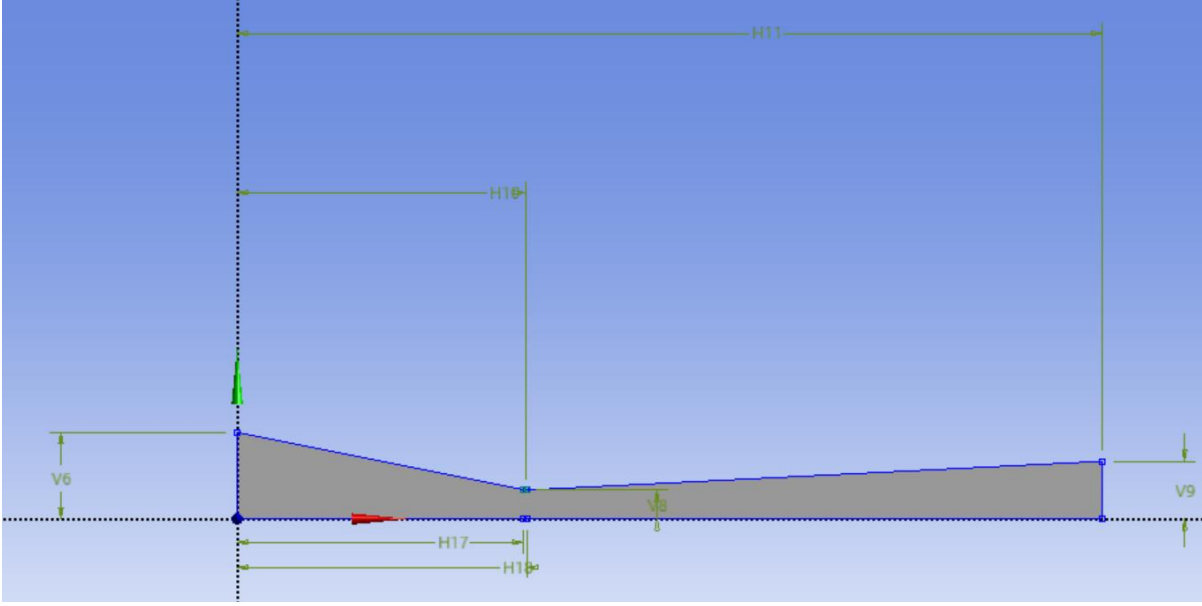


Figure 1: 2D Nozzle Geometry and Symmetry Boundary

2 Model Description

The numerical simulation of the supersonic nozzle is based upon the governing equations and conservation laws of fluid mechanics. To reduce the computational complexity of the system while maintaining physical accuracy, several assumptions were applied to the model. The flow is treated as a two dimensional, steady state, and inviscid compressible fluid. The inviscid assumption is used for this system because the primary focus of the analysis is the pressure and velocity gradients produced by the expansion geometry. In high speed nozzle flows, the Reynolds number is typically large enough that the influence of viscous boundary layers on the core flow field is negligible for validating the overall nozzle performance. Additionally, the working fluid is modeled as an ideal gas to simplify relationships. The first of these governing principles is the continuity equation which ensures the conservation of mass within the control volume and is as follows

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho v) = 0$$

In this equation, ρ represents the fluid density and v represents the velocity vector. The conservation of momentum is handled through the Euler equations which describe the relationship between the fluid acceleration and the pressure gradients without the influence of shear stress and are written as

$$\frac{\partial}{\partial t}(\rho v) + \nabla \cdot (\rho v v + pI) = 0$$

Here, p represents the static pressure and I represents the identity tensor. Finally, the conservation of energy is required to account for the temperature changes associated with the high speed expansion and is defined by the following expression

$$\frac{\partial}{\partial t}(\rho E) + \nabla \cdot (v(\rho E + p)) = 0$$

The variable E in this energy equation represents the total energy per unit mass. These equations are solved using a density based implicit solver in Fluent because it is specifically designed to handle the coupling between density, pressure, and temperature found in high speed compressible flows. This approach is appropriate for accurately capturing the expansion wave structures and potential shocks that are within supersonic nozzles.

3 Mesh

The generation of the mesh began with the creation of the nozzle geometry which is a half nozzle configuration with symmetry to reduce the total cell count. A structured quadrilateral mesh was selected for this study because it provides better alignment with the main flow direction compared to an unstructured triangular mesh. This choice is particularly important for supersonic flows where aligned grids minimize numerical diffusion across expansion waves and high speed gradients. To verify that the results were independent of the grid size, a mesh refinement process was implemented using three different levels of resolution. The refinement began with a coarse mesh of 1,250 cells and progressed to a medium mesh of 5,000 cells before reaching a high resolution fine mesh of 20,000 cells. Special attention was given to the throat region where the flow accelerates rapidly toward the sonic condition. This area required a significantly higher concentration of cells to resolve the steep pressure and velocity gradients as the fluid transitions from subsonic to supersonic speeds.

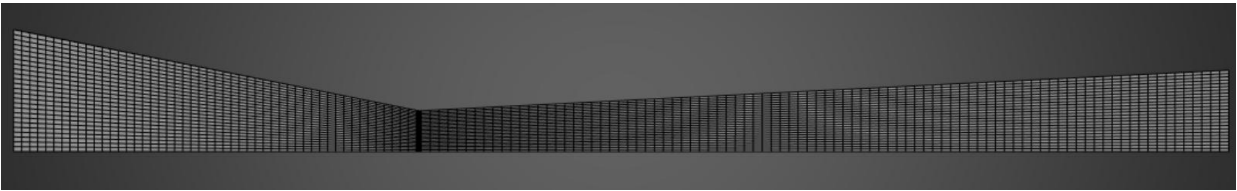


Figure 2: Structured Quadrilateral Mesh Layout for the Half-Nozzle Domain

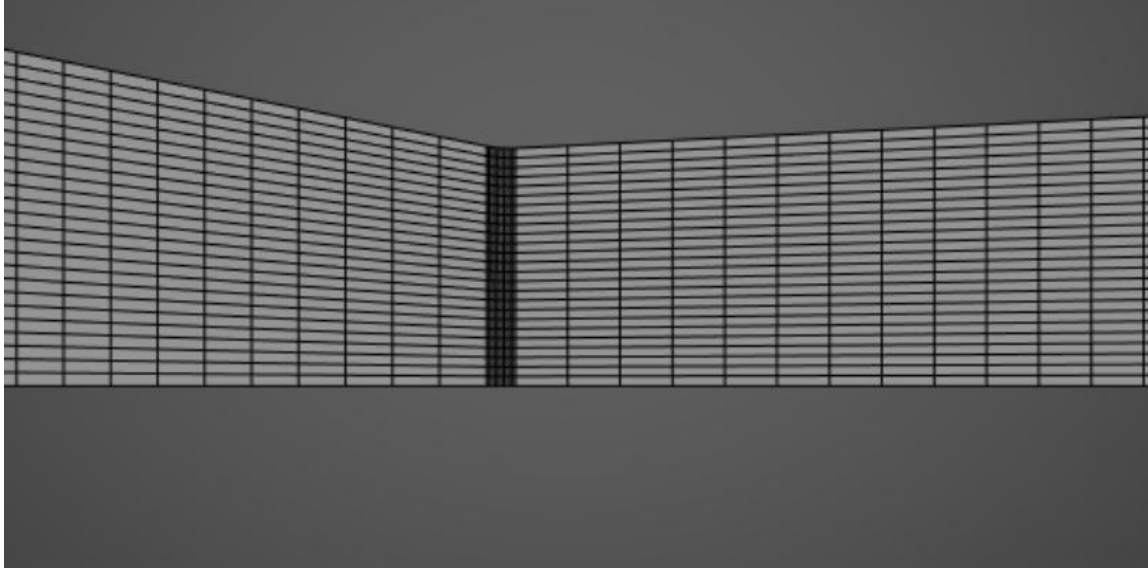


Figure 3: Close-up of Mesh Refinement and Cell Density at the Throat

4 Solution Methods

The numerical solution process utilized a density based implicit formulation within Fluent. This solver type is effective for high speed compressible flows because it couples the governing equations for mass, momentum, and energy into a single system. To handle the convective terms in the Euler equations, the Roe flux difference splitting scheme was selected capture accurate expansion wave gradients. For spatial discretization, a second order upwind scheme was applied to the pressure, density, momentum, and energy equations. This higher order discretization is necessary to minimize numerical diffusion and maintain the sharp gradients required for the validation.

Convergence of the numerical solution was monitored through several metrics to ensure the stability and accuracy of the results. One convergence criteria was based on the scaled residuals for each of the governing equations. These residuals were required to drop below a specific threshold before the simulation was considered converged. The target residual value for this study was defined as 1×10^{-6} . In addition to residual monitoring, integrated quantities such as the mass flow rate at the inlet and outlet were tracked to ensure the solution reached a steady state. To maintain stability during the initialization and the development of the supersonic flow field, modifications were made to the default relaxation parameters. Specifically, the Courant number was monitored as the expansion waves stabilized.

5 Verification

The verification phase of this analysis ensures that the numerical governing equations are being solved correctly and that the results are not dependent on the specific resolution of the

computational grid. To achieve this level of certainty, a grid independence study was performed by evaluating the exit Mach number across three distinct refinement levels. This process identifies the point at which further mesh refinement no longer significantly alters the solution, thereby confirming that the discretization error has been minimized. The results of this grid independence study for the coarse, medium, and fine mesh configurations are organized in the following table.

Table 1: Mesh Independence Study Results for Exit Mach Number

Refinement	Number of Cells	Exit Mach Number
Course	1,250	2.191
Medium	5,000	2.203
Fine	20,000	2.204

The data indicates that as the cell count increased from 1,250 cells in the coarse case to 5,000 cells in the medium case, the exit Mach number shifted from 2.191 to 2.203. Upon further refinement to the fine mesh of 20,000 cells, the Mach number reached a value of 2.204. To determine if the solution was truly independent of the grid, the relative percentage difference between the medium and fine cases was calculated using the following expression

$$Error = \frac{|M_{fine} - M_{medium}|}{M_{fine}} \times 100\%$$

The resulting difference between these two mesh levels was only 0.06 percent, which satisfies the requirement for a verified numerical solution. This extremely low variation confirms that the fine mesh of 20,000 cells is sufficient for the remainder of the analysis and that any remaining errors are likely due to physical modeling assumptions rather than numerical discretization.

6 Validation

The validation of the computational model involves a comparison between the numerical results and the published analytical solutions derived from one dimensional isentropic flow theory. This process is used to confirm that the physical assumptions and boundary conditions applied in the solver represent the gas dynamics of a supersonic expansion. For this study, the validation is performed across three critical performance parameters which are the exit Mach number, the conservation of mass flow, and the total net thrust.

The primary metric for validation is the Mach number at the nozzle exit which is determined by the area ratio of the geometry. According to isentropic tables for a ratio of 2.0, the target Mach number is 2.20. The computational results from the fine mesh yielded an exit Mach number of 2.204. This results in a numerical error of only 0.2 percent, providing strong evidence that the density based solver is correctly predicting the acceleration of the fluid through the diverging section.

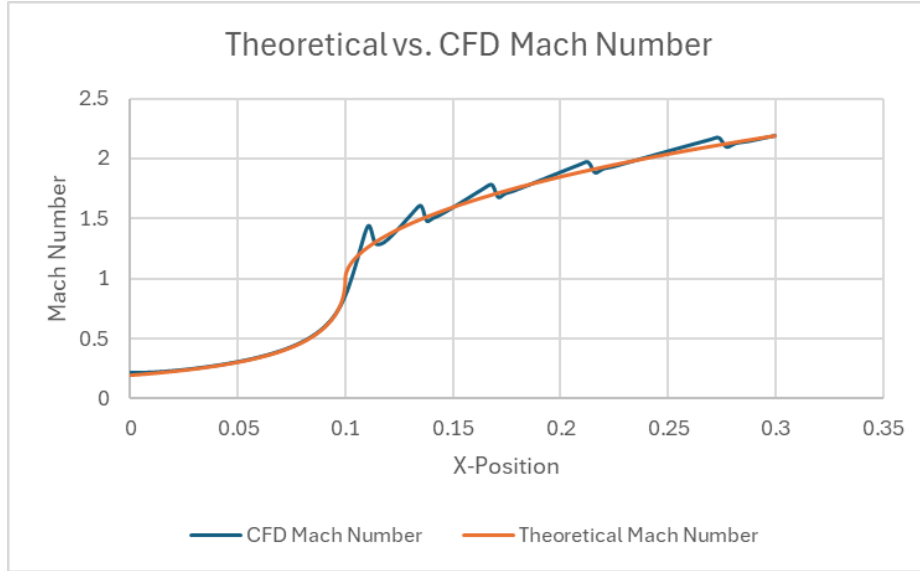


Figure 4: Comparison of Numerical Mach Number to 1D Isentropic Theory

A key observation in the comparison of Mach number distributions is the presence of minor periodic oscillations in the CFD results following the nozzle throat. While the one-dimensional isentropic theory assumes a perfectly uniform flow across any given cross-section, the two-dimensional computational model resolves the internal wave structures characteristic of supersonic expansion. These oscillations represent the propagation and reflection of expansion waves between the nozzle wall and the symmetry plane. As these waves interact with the flow field, they cause localized variations in velocity that appear to 'bounce' around the idealized theoretical curve. The close alignment of the numerical mean with the theoretical path, despite these fluctuations, confirms that the solver is correctly resolving the expansion physics while providing a more physically accurate representation of the internal environment than simplified analytical models.

The second validation point is the conservation of mass flow through the nozzle under choked conditions. The theoretical maximum mass flow rate is governed by the stagnation properties and the minimum area at the throat as defined by the following equation

$$\dot{m} = \frac{A^* P_0}{\sqrt{T_0}} \sqrt{\frac{\gamma}{R} \left(\frac{2}{\gamma + 1} \right)^{\frac{\gamma+1}{\gamma-1}}}$$

Based on the input conditions, the theoretical mass flow rate is 11.714 kilograms per second. The numerical report generated by the solver showed a mass flow rate of 11.714024 kilograms per second with a net flux error of nearly zero. This perfect agreement validates that the solver is accurately conserving mass as required by the continuity equation.

Table 2: Summary of Theoretical and Computational Validation Data

Performance Parameter	Theoretical Value	Computational Value	Percent Error
Exit Mach	2.2	2.204	0.2%
Mass Flow Rate	11.714 kg/s	11.714 kg/s	0%
Total Net Thrust	5,358 N	5,297 N	1.1%

Finally, the total net thrust produced by the nozzle was validated by accounting for both the momentum of the exiting gas and the pressure difference at the exit plane. The total thrust was calculated using the following relationship

$$F = \dot{m}V_e + (P_e - P_a)A_e$$

The theoretical calculation for these conditions resulted in a thrust of 5,358 N. The numerical data from the simulation produced a net thrust of 5,297 N, which represents a difference of approximately 1.1 percent. This minor deviation is consistent with the presence of two dimensional wave structures in the CFD model that are not accounted for in simplified one dimensional theory. These three validation points collectively confirm that the computational model is a highly accurate representation of the nozzle performance.

7 Results and Discussion

The results of the CFD analysis provide a visualization of the supersonic expansion process within the nozzle. The static pressure contours show a continuous decrease from the inlet stagnation pressure toward the exit plane as the fluid accelerates through the converging and diverging sections. This acceleration is clearly depicted in the Mach number contours where the flow transitions from a subsonic state to the sonic state exactly at the throat. The development of the supersonic flow field is characterized by the presence of expansion waves that originate at the throat corners and propagate through the diverging section. These wave structures are a direct consequence of the two dimensional nature of the flow and represent a key distinction from simplified one dimensional analytical models.

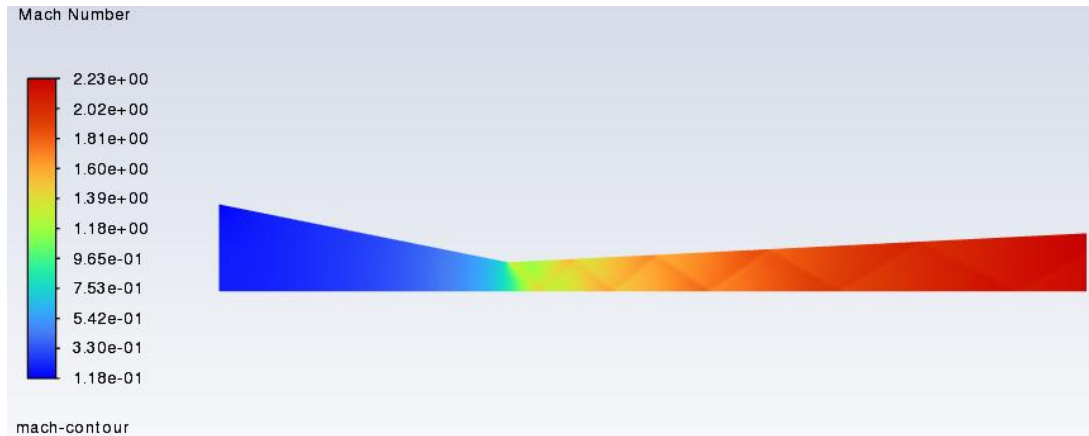


Figure 5: Mach Number Contour Distribution across the Nozzle Domain

A critical feature identified in the results is the over expanded state of the nozzle at the design ambient pressure of 101,325 Pascals. The numerical analysis shows that the static pressure at the exit plane reaches approximately 47,015 Pascals which is significantly lower than the atmospheric pressure. This pressure differential causes the exhaust gases to be compressed by the surrounding atmosphere as they leave the nozzle which physically explains the negative pressure thrust component identified in the validation section. Despite this over expansion, the velocity vectors remain highly aligned with the nozzle axis which confirms that the majority of the fluid energy is successfully converted into axial momentum.

The comparison between the CFD centerline Mach number and the isentropic curve reveals minor periodic oscillations in the numerical data. These fluctuations are caused by the reflection of expansion waves between the inviscid wall and the symmetry line of the domain. While isentropic theory assumes a perfectly uniform flow across any given cross section, the CFD model captures these internal wave interactions which are characteristic of real world nozzles. These features prove that the computational model is not only accurate in its integrated results but also captures the internal physics of the supersonic flow field.

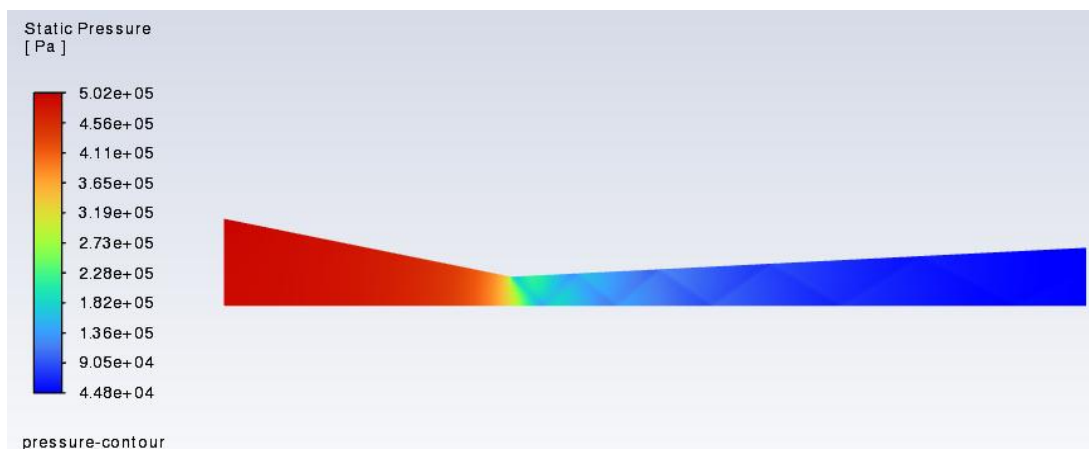


Figure 6: Static Pressure Contour Map from Inlet to Exit

8 Conclusion

The computational analysis of the supersonic converging-diverging nozzle has successfully satisfied the objectives established at the beginning of this study. Through the implementation of a density based implicit solver and a structured quadrilateral mesh, the simulation provided a highly accurate representation of the internal gas dynamics. The verification process confirmed that the results were independent of the grid resolution, as the fine mesh reached a stable exit Mach number with a negligible variation of only 0.06 percent compared to the medium mesh. This numerical certainty allowed for a solid validation against one dimensional isentropic flow theory.

The final validation results demonstrated that the computational model is highly accurate for predicting nozzle performance parameters. The exit Mach number of 2.204 and the mass flow rate of 11.714 kilograms per second both showed great agreement with theoretical targets, maintaining errors below 0.2 percent. Additionally, the calculated net thrust of 5,297.09 Newtons was within 1.1 percent of the analytical prediction. These results confirm that the chosen solver configuration is appropriate for capturing the momentum flux and pressure distributions in supersonic regimes.

Beyond the numerical values, the simulation successfully captured the critical physical features of the flow field. The Mach contours verified that the flow reached the sonic condition precisely at the throat and continued to accelerate through a series of expansion waves in the diverging section. The analysis also correctly identified the over expanded state of the nozzle, where the exit pressure fell below the ambient atmospheric pressure. By resolving these two dimensional wave interactions and pressure differentials, the study concludes that the density based CFD approach provides a reliable and verified method for the design and analysis of supersonic propulsion systems.

References

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- [2] Ansys Fluent Theory Guide (2025 R2). *Section 23.1.2: Density-Based Solver*.
- [3] CFDLAND (2024). *Compressible Flow in a Nozzle: Step-by-Step Tutorial Guide In ANSYS Fluent*.
- [4] [Gas tables_v2.pdf](#)
- [5] Versteeg, H. K., & Malalasekera, W. (2007). *An Introduction to Computational Fluid Dynamics: The Finite Volume Method*. Pearson Education.

Appendix

Table A.1: Isentropic flow properties for $\gamma = 1.4$

M	T/T_0	P/P_0	A/A^*	PA/P_0A^*	μ	ν
0.02	0.9999	0.9997	28.9421	28.9340	-	-
0.04	0.9997	0.9989	14.4815	14.4653	-	-
0.06	0.9993	0.9975	9.6659	9.6416	-	-
0.08	0.9987	0.9955	7.2616	7.2292	-	-
0.10	0.9980	0.9930	5.8218	5.7813	-	-
-	-	-	-	-	-	-
0.12	0.9971	0.9900	4.8643	4.8156	-	-
0.14	0.9961	0.9864	4.1824	4.1255	-	-
0.16	0.9949	0.9823	3.6727	3.6077	-	-
0.18	0.9936	0.9776	3.2779	3.2047	-	-
0.20	0.9921	0.9725	2.9635	2.8820	-	-
-	-	-	-	-	-	-
0.22	0.9904	0.9668	2.7076	2.6178	-	-
0.24	0.9886	0.9607	2.4956	2.3975	-	-
0.26	0.9867	0.9541	2.3173	2.2109	-	-
0.28	0.9846	0.9470	2.1656	2.0508	-	-
0.30	0.9823	0.9395	2.0351	1.9119	-	-
-	-	-	-	-	-	-
0.32	0.9799	0.9315	1.9219	1.7902	-	-
0.34	0.9774	0.9231	1.8229	1.6827	-	-
0.36	0.9747	0.9143	1.7358	1.5871	-	-
0.38	0.9719	0.9052	1.6587	1.5014	-	-
0.40	0.9690	0.8956	1.5901	1.4242	-	-
-	-	-	-	-	-	-
0.42	0.9659	0.8857	1.5289	1.3542	-	-
0.44	0.9627	0.8755	1.4740	1.2905	-	-
0.46	0.9594	0.8650	1.4246	1.2322	-	-
0.48	0.9559	0.8541	1.3801	1.1788	-	-
0.50	0.9524	0.8430	1.3398	1.1295	-	-
-	-	-	-	-	-	-
0.52	0.9487	0.8317	1.3034	1.0840	-	-
0.54	0.9449	0.8201	1.2703	1.0417	-	-
0.56	0.9410	0.8082	1.2403	1.0024	-	-
0.58	0.9370	0.7962	1.2130	0.9658	-	-
0.60	0.9328	0.7840	1.1882	0.9316	-	-
-	-	-	-	-	-	-
0.62	0.9286	0.7716	1.1656	0.8995	-	-
0.64	0.9243	0.7591	1.1451	0.8693	-	-
0.66	0.9199	0.7465	1.1265	0.8410	-	-
0.68	0.9153	0.7338	1.1097	0.8142	-	-
0.70	0.9107	0.7209	1.0944	0.7890	-	-
-	-	-	-	-	-	-
0.72	0.9061	0.7080	1.0806	0.7651	-	-
0.74	0.9013	0.6951	1.0681	0.7424	-	-
0.76	0.8964	0.6821	1.0570	0.7209	-	-
0.78	0.8915	0.6691	1.0471	0.7005	-	-
0.80	0.8865	0.6560	1.0382	0.6811	-	-

M	T/T_0	P/P_0	A/A^*	PA/P_0A^*	μ	ν
0.82	0.8815	0.6430	1.0305	0.6626	-	-
0.84	0.8763	0.6300	1.0237	0.6449	-	-
0.86	0.8711	0.6170	1.0179	0.6281	-	-
0.88	0.8659	0.6041	1.0129	0.6119	-	-
0.90	0.8606	0.5913	1.0089	0.5965	-	-
-	-	-	-	-	-	-
0.92	0.8552	0.5785	1.0056	0.5817	-	-
0.94	0.8498	0.5658	1.0031	0.5675	-	-
0.96	0.8444	0.5532	1.0014	0.5539	-	-
0.98	0.8389	0.5407	1.0003	0.5409	-	-
1.00	0.8333	0.5283	1.0000	0.5283	90.000	0.000
-	-	-	-	-	-	-
1.02	0.8278	0.5160	1.0003	0.5162	78.635	0.126
1.04	0.8222	0.5039	1.0013	0.5045	74.058	0.351
1.06	0.8165	0.4919	1.0029	0.4933	70.630	0.637
1.08	0.8108	0.4800	1.0051	0.4825	67.808	0.968
1.10	0.8052	0.4684	1.0079	0.4721	65.380	1.336
-	-	-	-	-	-	-
1.12	0.7994	0.4568	1.0113	0.4620	63.234	1.735
1.14	0.7937	0.4455	1.0153	0.4523	61.306	2.160
1.16	0.7879	0.4343	1.0198	0.4428	59.550	2.607
1.18	0.7822	0.4232	1.0248	0.4337	57.936	3.074
1.20	0.7764	0.4124	1.0304	0.4249	56.443	3.558
-	-	-	-	-	-	-
1.22	0.7706	0.4017	1.0366	0.4164	55.052	4.057
1.24	0.7648	0.3912	1.0432	0.4081	53.751	4.569
1.26	0.7590	0.3809	1.0504	0.4001	52.528	5.093
1.28	0.7532	0.3708	1.0581	0.3924	51.375	5.627
1.30	0.7474	0.3609	1.0663	0.3848	50.285	6.170
-	-	-	-	-	-	-
1.32	0.7416	0.3512	1.0750	0.3775	49.251	6.721
1.34	0.7358	0.3417	1.0842	0.3704	48.268	7.279
1.36	0.7300	0.3323	1.0940	0.3636	47.332	7.844
1.38	0.7242	0.3232	1.1042	0.3569	46.439	8.413
1.40	0.7184	0.3142	1.1149	0.3504	45.585	8.987
-	-	-	-	-	-	-
1.42	0.7126	0.3055	1.1262	0.3440	44.767	9.565
1.44	0.7069	0.2969	1.1379	0.3379	43.983	10.146
1.46	0.7011	0.2886	1.1501	0.3319	43.230	10.731
1.48	0.6954	0.2804	1.1629	0.3261	42.507	11.317
1.50	0.6897	0.2724	1.1762	0.3204	41.810	11.905
-	-	-	-	-	-	-
1.52	0.6840	0.2646	1.1899	0.3149	41.140	12.495
1.54	0.6783	0.2570	1.2042	0.3095	40.493	13.086
1.56	0.6726	0.2496	1.2190	0.3042	39.868	13.677
1.58	0.6670	0.2423	1.2344	0.2991	39.265	14.269
1.60	0.6614	0.2353	1.2502	0.2941	38.682	14.860

M	T/T_0	P/P_0	A/A^*	PA/P_0A^*	μ	ν
1.62	0.6558	0.2284	1.2666	0.2893	38.118	15.452
1.64	0.6502	0.2217	1.2836	0.2845	37.572	16.043
1.66	0.6447	0.2151	1.3010	0.2799	37.043	16.633
1.68	0.6392	0.2088	1.3190	0.2754	36.530	17.222
1.70	0.6337	0.2026	1.3376	0.2710	36.032	17.810
-	-	-	-	-	-	-
1.72	0.6283	0.1966	1.3567	0.2667	35.549	18.396
1.74	0.6229	0.1907	1.3764	0.2625	35.080	18.981
1.76	0.6175	0.1850	1.3967	0.2584	34.624	19.565
1.78	0.6121	0.1794	1.4175	0.2544	34.180	20.146
1.80	0.6068	0.1740	1.4390	0.2504	33.749	20.725
-	-	-	-	-	-	-
1.82	0.6015	0.1688	1.4610	0.2466	33.329	21.302
1.84	0.5963	0.1637	1.4836	0.2429	32.921	21.877
1.86	0.5910	0.1587	1.5069	0.2392	32.523	22.449
1.88	0.5859	0.1539	1.5308	0.2356	32.135	23.019
1.90	0.5807	0.1492	1.5553	0.2321	31.757	23.586
-	-	-	-	-	-	-
1.92	0.5756	0.1447	1.5804	0.2287	31.388	24.151
1.94	0.5705	0.1403	1.6062	0.2253	31.028	24.712
1.96	0.5655	0.1360	1.6326	0.2220	30.677	25.271
1.98	0.5605	0.1318	1.6597	0.2188	30.335	25.827
2.00	0.5556	0.1278	1.6875	0.2157	30.000	26.380
-	-	-	-	-	-	-
2.02	0.5506	0.1239	1.7160	0.2126	29.673	26.930
2.04	0.5458	0.1201	1.7451	0.2096	29.353	27.476
2.06	0.5409	0.1164	1.7750	0.2066	29.041	28.020
2.08	0.5361	0.1128	1.8056	0.2037	28.736	28.560
2.10	0.5313	0.1094	1.8369	0.2009	28.437	29.097
-	-	-	-	-	-	-
2.12	0.5266	0.1060	1.8690	0.1981	28.145	29.631
2.14	0.5219	0.1027	1.9018	0.1954	27.859	30.161
2.16	0.5173	0.0996	1.9354	0.1927	27.578	30.688
2.18	0.5127	0.0965	1.9698	0.1901	27.304	31.212
2.20	0.5081	0.0935	2.0050	0.1875	27.036	31.732
-	-	-	-	-	-	-
2.22	0.5036	0.0906	2.0409	0.1850	26.773	32.249
2.24	0.4991	0.0878	2.0777	0.1825	26.515	32.763
2.26	0.4947	0.0851	2.1153	0.1801	26.262	33.273
2.28	0.4903	0.0825	2.1538	0.1777	26.014	33.780
2.30	0.4859	0.0800	2.1931	0.1754	25.771	34.283
-	-	-	-	-	-	-
2.32	0.4816	0.0775	2.2333	0.1731	25.533	34.782
2.34	0.4773	0.0751	2.2744	0.1709	25.300	35.279
2.36	0.4731	0.0728	2.3164	0.1687	25.070	35.771
2.38	0.4688	0.0706	2.3593	0.1665	24.845	36.261
2.40	0.4647	0.0684	2.4031	0.1644	24.624	36.747